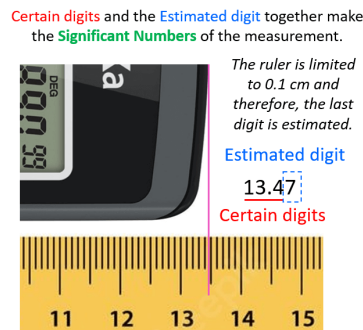


Whenever we make a **measurement**, there is always going to be some **uncertainty** because of the precision that the instrument allows. For example, let's say we measure the length of a ruler.

We can all agree that it is longer than 13.4 (certain digits) but, we cannot say for sure what the digit after the 4 is. Therefore, we estimate it. Let's say we may estimate that the length is 13.47 cm. All these numbers together, that is the **certain digits** (13.4) and one **estimated digit** (7), make the **significant figures** of the measurement or simply significant numbers.



This is important because when making a measurement or performing mathematical operations, we must **report the answer based on the certainty of our measurement** or the given number(s). Therefore, the next question is:

How do I Identify the Significant Figures in a Number?

The first rule is simple, any **nonzero number is a significant figure**. For example, 26 has **two S.F.**, 164 has **three S.F.**, 18948 has **five S.F.** and etc.

All nonzero digits are significant figures.

26	844	59841	656241
2 S.F.	3 S.F.	5 S.F.	6 S.F.

For the zeros in different positions, here is what you need to remember:

The zeros in between nonzero digits (**interior zeroes**) are considered **significant figures**.

For example, 6204 has **4 S.F.**, 84069 has **5 S.F.**, and 204 has **3 S.F.**

Interior zeroes (zeroes between two nonzero digits) are significant.

206	804	5041	506041
3 S.F.	3 S.F.	4 S.F.	6 S.F.

If the zeros are at the end of the number (**trailing zeroes**), they are considered **nonsignificant**, unless there is a decimal place!

Zeros at the End of the Number

210	6500	15460	598440	2.10	6.500	1.5460	5.98440
2 S.F.	2 S.F.	4 S.F.	5 S.F.	3 S.F.	4 S.F.	5 S.F.	6 S.F.

Trailing zeroes **without a decimal point** are **not significant**.

Trailing zeroes **after a decimal point** are always **significant**.

The reason for this is that the *zeros without a decimal point do not add any certainty* to the measurement, and only increases the number. Remember, **significant figures are about the certainty of the measurement** and not just any digit that appears in the number.

The same goes for zeros that stand before the first nonzero digit (**leading zeroes**). These are not significant as they only tell us how small the number is and not how accurate the measurement was.

Leading zeroes (zeroes before the first nonzero digit) are not significant.

0.75	0.00681	0.05084	0.052680
2 S.F.	3 S.F.	4 S.F.	5 S.F.

If the *trailing zero appears before the decimal point with following zeroes*, then it is also a **significant number**:

Trailing zeros before the decimal point with following zeroes, are also significant number:

260.40	6840.000	150.0	50.9040
5 S.F.	7 S.F.	4 S.F.	6 S.F.

Significant Figures and Scientific Notation

For counting the number of significant figures in scientific notation, we **focus on the N**, that is the **number that comes before the 10ⁿ**. The exponent is only to tell us about the magnitude of the number.

N x 10ⁿ

Only this **counts for significant figures**.

1.34 x 10 ⁵	1.34000 x 10 ⁵	8.6 x 10 ⁻⁴
3 S.F.	6 S.F.	2 S.F.

Exact Numbers

These are the numbers that **do not have any uncertainty** and therefore, they do not limit the number of significant figures.

For example, there are 20 students in a class, and that number is an exact number. It is not like we can argue that it is 20.1 or 20.3 depending on how we look at it. In other words, we do not use an instrument with certain precision to measure these numbers. Therefore, they are called **exact numbers** and they **have an infinite number of significant figures**.

Some more examples: How many significant figures are in each number?

a) 632, b) 45.0, c) 4706.0, d) 0.0075, e) 0.006840, f) 0.15000, g) 2.30 x 10⁴, h) 5,000

a) 632, b) 45.0, c) 4706.0, d) 0.0075, e) 0.006840, f) 0.15000, g) 2.30 x 10⁴, h) 5,000

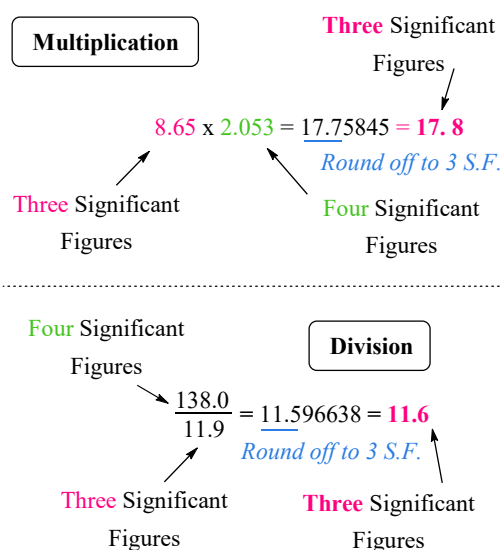
3 S.F. 3 S.F. 5 S.F. 2 S.F. 4 S.F. 5 S.F. 3 S.F. 1 S.F.

Significant Figures in Mathematical Operations

There are two main rules applied to **addition and subtraction** and **multiplication and division** respectively.

Significant Figures in Multiplication and Division

The answer should have the same number of significant figures as the factor with the fewest significant figures.



Significant Figures in Addition and Subtraction

When doing **addition or subtraction**, the **answer** carries the **same number of decimal places** as the number with the **fewest decimal places**.

$$\begin{array}{r} 4.4981 \text{ Four decimal places} \\ - 1.26 \text{ Two decimal places} \\ \hline 3.2381 \end{array} \rightarrow 3.24$$

Round off to two decimal places

What if one or all the numbers do not have decimal places?

Remember, the whole idea of rounding off is to drop the insignificant digits. These are all the digits to the right of the last significant digit.

The last (only) significant figure has a certainty of hundreds which is less than tens and ones of 5289.

$5,389 \approx 5,400$ or 5.4×10^3

Therefore, the answer is rounded to hundreds.

Significant Figures

Significant Numbers in Multistep Calculations

If there are more than two numbers involved in the calculation, the rule is to **not round off any intermediate result** and only do it for the final answer.

$$3.54 \times 2.6458 \times 7.48764 = 70.130224 \approx 70.1$$

3 S.F. 5 S.F. 5 S.F. 3 S.F.

Multistep Calculations with Mixed Operations

Multistep calculations involving both subtraction division and addition subtraction

$$4.647 \times 2.5149 \times (6.2594 - 2.34)$$

4 D.P. 2 D.P.
 Parenthesis first

Do not round off yet.

Multiply by the product of 4.647 and 2.5149 first and only then round off the answer.

$$3.9194$$

The last decimal indicating that in the next steps this number should be considered as 3 S.F.

$$4.647 \times 2.5149 \times 3.9194 = 45.8$$

3 S.F. 3 S.F.

Another example

$$\frac{2.487 \times 3.1629}{4.1926 - 2.07} = \frac{2.487 \times 3.1629}{2.1226} = 3.7058947 \approx 3.71$$

2.1226 2.1226 3 S.F.

Two decimal places means this number will be considered having 3 S.F. and because this is fewer than 2.487 and 3.1629, the final answer should have 3 S.F.

$$\frac{2.487 \times 3.1629}{2.1226} = \frac{7.8661323}{2.1226} = 3.7058947 \approx 3.71$$

2.1226 2.1226 3 S.F.

Significant Figures in Calculations with Scientific Notation

If the **exponents on the 10s are the same**, you can just add the numbers in front of the powers of ten and put the exponent next to the answer:

$$2.65 \times 10^3 + 6.4 \times 10^3 = ?$$

1 D.P.

$$2.65 + 8.4 = 9.05 \approx 9.1 \times 10^3$$

1 D.P.

If the **exponents are not identical**, then we can **take both numbers out of scientific notation**, do the addition or subtraction, and then put the answer back into scientific notation. For example,

$$9.578 \times 10^3 - 5.326 \times 10^2 = ?$$

$$9578 - 532.6 = 9,045.4 = 9.045 \times 10^3$$

We can still draw the line after the last significant figure to make this more visual:

Scientific Notation in Multiplication and Division

When **multiplying** the **exponents are added**, and when **dividing**, the **exponents are subtracted**.

$$10^x \times 10^y = 10^{x+y}$$

$$10^x \div 10^y = 10^{x-y}$$

The numbers in front of the 10s are multiplied and divided accordingly.

Add

$$6 \times 10^x \times 4 \times 10^y = 24 \times 10^{x+y}$$

Multiply

Subtract

$$8 \times 10^x \div 4 \times 10^y = 2 \times 10^{x-y}$$

Divide

Example: $3,400.00 \times 0.624 + 230 = ?$

The first step multiplication gives 2,121.6. **Do not round off yet** - just **keep track** of two significant figures and do the addition.

$$3,400.00 \times 0.624 = 2,121.6$$

2 S.F. 2 S.F.
 Don't round yet

$$\begin{array}{r} 2,121.6 \\ + 230 \\ \hline 2,351.6 \end{array} \rightarrow 2,400$$

Round up the 3